AWS User Groups Ahmedabad

acm Chapter Event Ahmedabad

AHMEDABAD MANAGEMENT ASSOCIATION

COMPUTER SOCIETY OF INDIA ESTD. 1965 सर्वे भवन्तु सुखिनः

aws

COMMUNITY DAY

AHMEDABA

DEVOPS D EDITION

10th Oct, 2026



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

Watching Self-Repair Happen

Automatic Detection and Recovery on ECS Managed Instances.



Avinash Dalvi

Technical Architect @Quantiphi

AWS UG Bengaluru Co-Leader & Serverless Hero



This wasn’t my client ask or my sprint work. As an Architect and product builder, one question keep pulling at me

When GPU actually dies,

What does ECS really do ?

AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

To answer that, first see how failure looks to the platform.



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

First case: a burner dies, everyone sees it.



Second case: the tandoor looks hot and the naan comes out raw. That is what a GPU failure looks like to ECS.

AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

GPUs will fails. Plan for it

- A memory error (ECC)
- A bad connection between GPUs (NvLink)
- A card that drops off the bus.

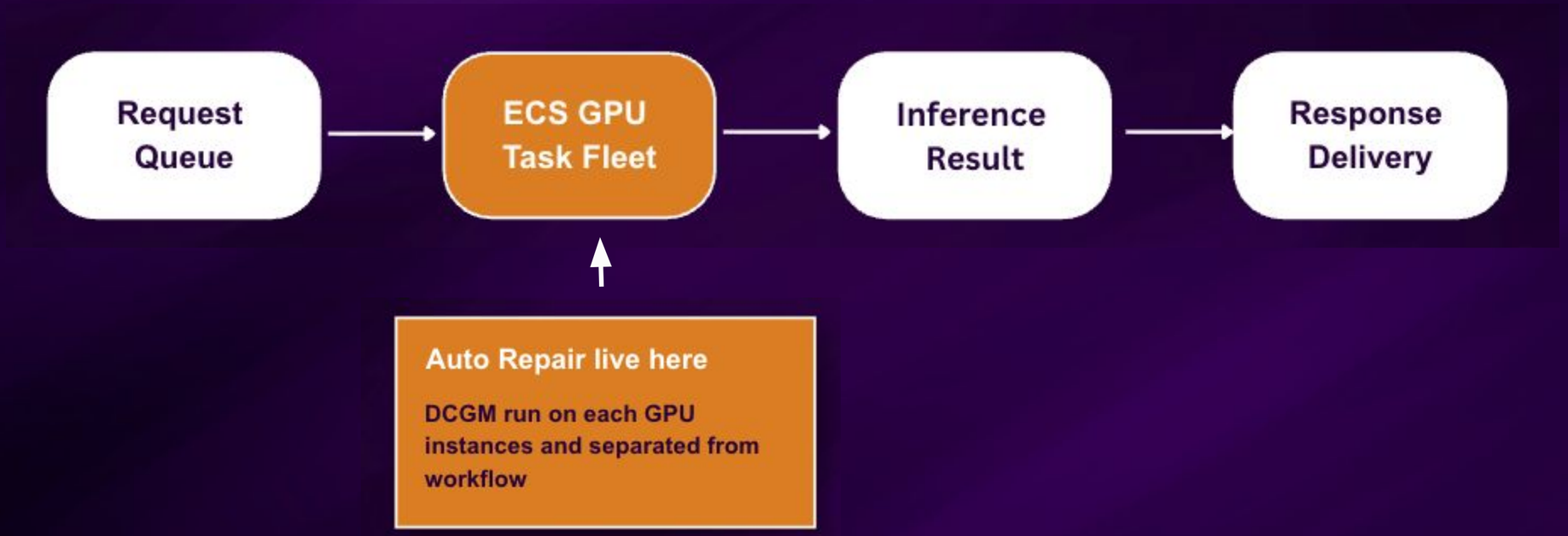
Today, on self managed fleet

1. Health check script poll NVIDIA -SMI
2. CloudWatch alarm fires
3. Lambda deregister for instance
4. Manual or scripted instance refresh

Every hop is a new place to fail.



Where auto-repair fits in pipeline



What actually I did

Wanted to see what changes with ECS managed instances.

1

Setup AWS sample

Sample project from AWS - sample-ecs-inference-pipeline

2

Faked GPU Error

Told GPU monitor that error 79 happened. Didn't break any hardware

3

Watched

Didn't click or run anything after that





Injecting a GPU fault.

```
dcgmi test --inject -f 230 -v 79
```

230 = XID error field, 79 = GPU fell off the bus.

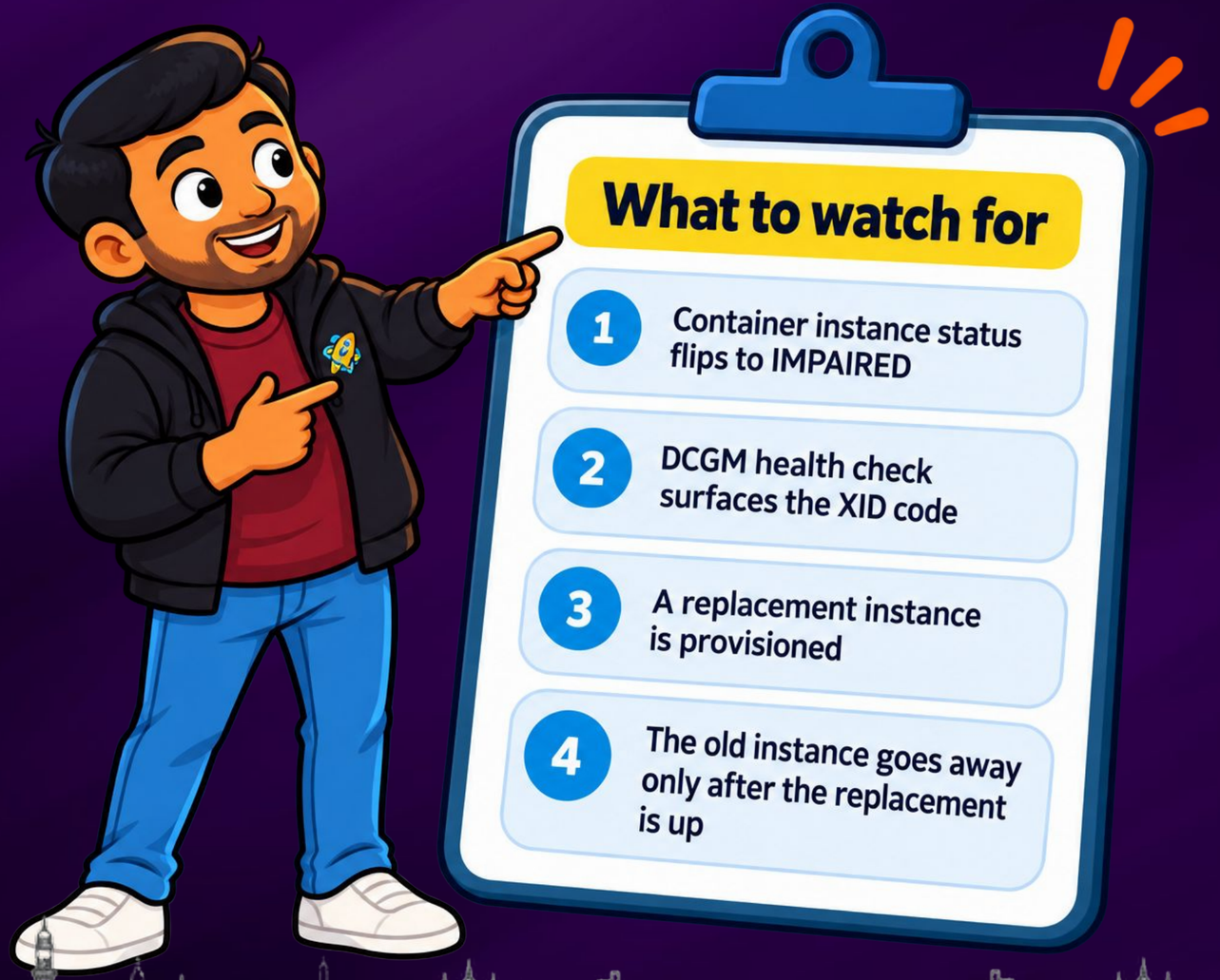
My setup looks like

- 1 GPU instance g6.2xlarge, NVIDIA L4
- 1 inference task running on it
- Fault injected from privileged task on same instance
- The commands only change what DCGM reports, ECS reacts to that report



AWS User Groups
Ahmedabad

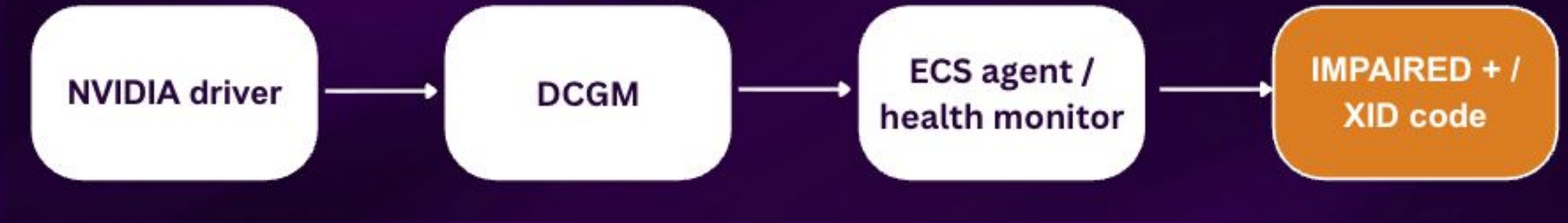
aws
COMMUNITY DAY
AHMEDABA
DEVOPS EDITION



What to watch for

1. Container instance status flips to IMPAIRED
2. DCGM health check surfaces the XID code
3. A replacement instance is provisioned
4. The old instance goes away only after the replacement is up

How the fault gets noticed



This whole path normally runs silently in the background — that's the point.

AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

EC2 says ok. ECS already knows.

In my runs, EC2 still said ok 2 and 6 minutes after the fault was flagged

```
INSTANCES: 2
-------------------------------------------------------------------------------------
|                                DescribeContainerInstances                         |
+----------+-------------------------+-----------+---------+----------+-------------+
|  accel   |          ec2            |  overall  | reason  | running  |   state     |
+----------+-------------------------+-----------+---------+----------+-------------+
|  OK      |  i-007442553fab7f316    |  OK       |  None   |  0       |  ACTIVE     |
|  IMPAIRED|  i-0eef88b856bb16fb3    |  IMPAIRED |  XID_79 |  1       |  DRAINING   |
+----------+-------------------------+-----------+---------+----------+-------------+
zsh: command not found: #
OLD EC2 (i-0eef88b856bb16fb3): ok       running ok
```

The instance looked healthy to EC2 while ECS already knew about the GPU fault. That is the tandoor.



How you find out: EventBridge

```
{
  "detail-type": "ECS Container Instance Health Change",
  "detail": {
    "ec2InstanceId": "i-0...",
    "overallStatus": "IMPAIRED",
    "healthChecks": [{
      "type": "ACCELERATED_COMPUTE",
      "status": "IMPAIRED",
      "statusReason": "XID_79"
    }]
  }
}
```



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

Start-before-stop: replace first, then remove

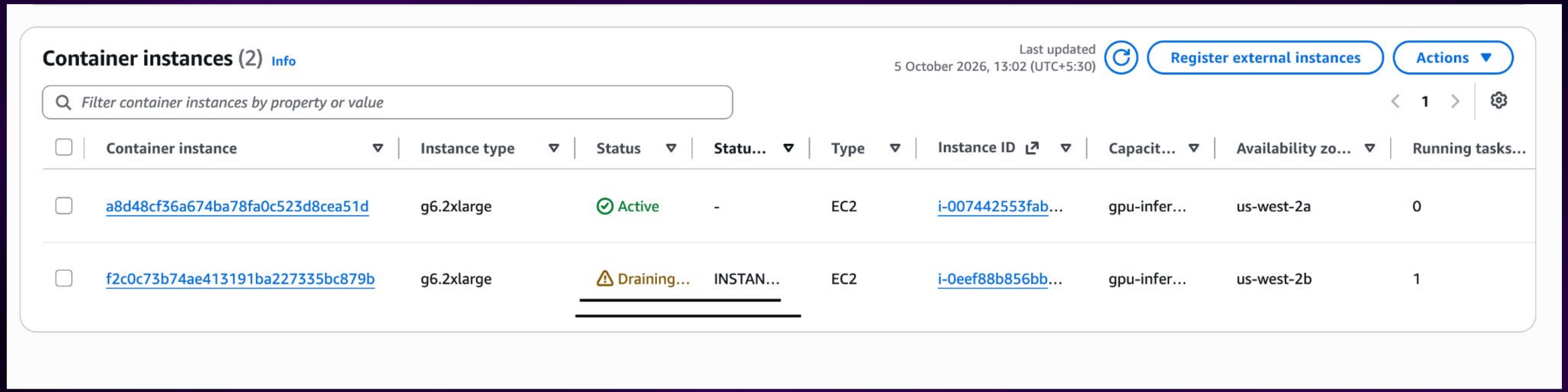
Detection takes seconds. The repair takes minutes, and the replacement comes first.



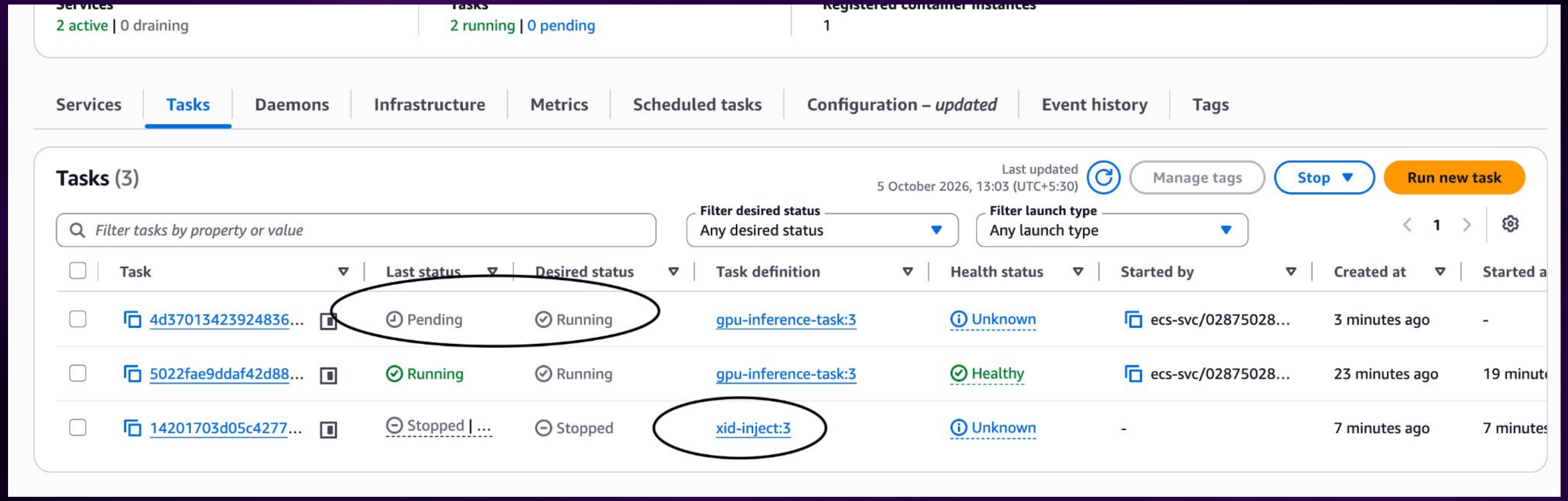
18 – 49 s	3.5 – 4.5 min	10.5 – 12 min
(to detect (three runs)	(until the replacement instance appears)	until the old task stops (1 → 2 → 1 instances)

Two instances at once

Detection takes seconds. The repair takes minutes, and the replacement comes first.



New task starts, old task stops



When repair can’t start

My first run, with the fleet sitting at its vCPU quota: 4 × g6.2xlarge = 32 vCPUs.

24 s

Fault detected. ECS flags the GPU as IMPAIRED.

DRAINING

The impaired instance stops taking work. Repair starts.

46 min

Still stuck. VcpuLimitExceeded kept firing in the service events.

Replace-before-remove needs a spare instance. A spare instance needs spare quota. Keep your vCPU quota above fleet size plus one instance.



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

Scale-to-zero isn’t zero

The mistake that gave me a bill I didn’t plan for.

What I did

Set the service's desired count to 0 and walked away.

What I missed

The scaling policy was still active. Work waiting in the queue kept pushing the desired count back up, and GPU instances kept launching.

What I check now

Pin max capacity to 0, or suspend scaling / Purge the queue so the alarm clears / List running instances. Don't trust the setting / Set a budget alert / Delete the stack when done

A setting that says zero is a request. A list of running instances is proof.



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

Before you run this in production

Tested	**Quota headroom**	Keep vCPU quota above fleet size plus one instance. Without it, my repair stalled.
Tested	**More than one instance type**	g6e had no capacity for me, so I allowed g6.2xlarge as a fallback. Repair worked on it.
Partly tested	**Alerts to on-call**	The event arrived about 60 s after the status change. The sample uses a log group and SNS. I did not test SNS.
Tested	**Cost guardrails**	Set a budget alert. Check running instances after teardown. Delete the stack when you are done.
Docs only	**Placement and IAM**	How GPU tasks sit next to non-GPU services, and which permissions auto-repair needs.

Not Tested: A physically fail GPU, Other XID code, request in flight during repair, the large GPU tier





AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

When GPU actually dies, what does ECS really do?

It notices in seconds, replaces first, removes later, if you have spare quota.



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

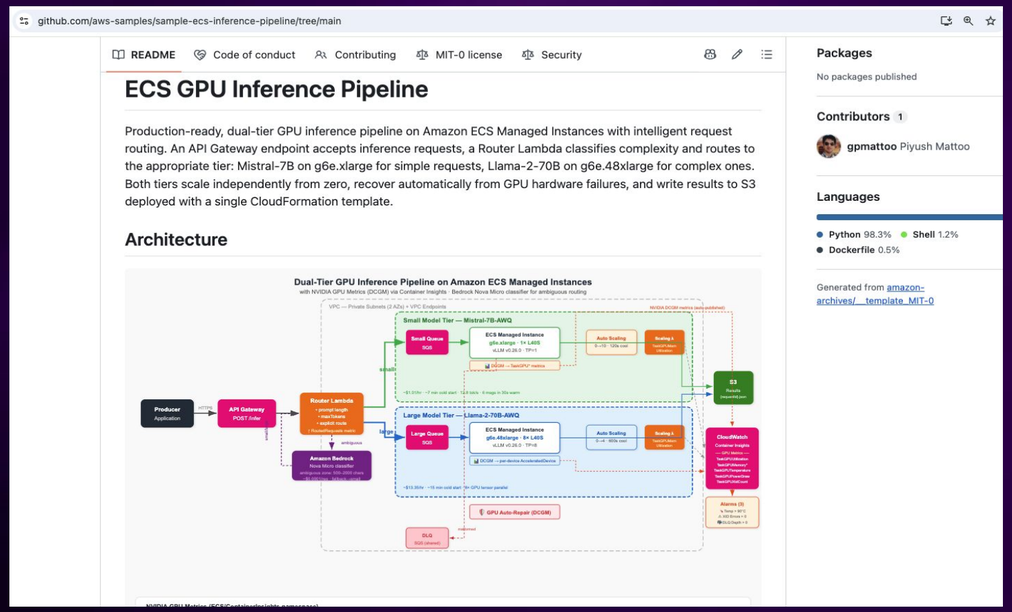
Is your vCPU quota above fleet size plus one?



AWS User Groups Ahmedabad

aws COMMUNITY DAY AHMEDABA DEVOPS EDITION

References



github.com/aws-samples/sample-ecs-inference-pipeline

AWS User Groups
Ahmedabad

aws
COMMUNITY DAY
AHMEDABA
DEVOPS EDITION



https://www.internetkatta.com/



AWS
User Groups
Ahmedabad

aws
COMMUNITY DAY
AHMEDABA
DEVOPS EDITION

THANK You !!

